No of Pages : 3 **Course Code : 23MX24**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, JUNE 2024

MCA Semester : 2

23MX24 ENTERPRISE COMPUTING USING FULL STACK

Time : 3 Hours **Maximum Marks :100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. For all questions, Sub division A takes three marks, B takes 10 marks and C takes twelve marks.
3. Course Outcome : Table

Qn.1	CO1	Qn.2	CO2	Qn.3	CO3	Qn.4	CO4

COs	At the end of learning this course, students will be able to
CO1	Exemplify the concepts of Enterprise, Enterprise architecture and applications and using Java and Java Script Technology Patterns and Frameworks
CO2	Develop enterprise Web Database Application using SQL, NOSQL and JAVA Frameworks
CO3	Develop Enterprise Application using Java script based Frameworks
CO4	Develop inter-enterprise applications using SOAP and REST based Web Services and Micro Services

		Question	(BTL)
1	a)	In what way does Separation of Concerns in Java and Java Script based Full Stack simplify the enterprise application developments? (3)	(L4)
	b)	Perform a critical assessment of Java Full stack and any one Java Script based Software Stack Technologies and its Methodologies to be adopted in the development of the following case study of an Enterprise Application. Develop a Web based system for making on-line purchases. This would include presenting lists of items, selecting items (verifying that the items exist and are in stock), totalling up and computing a bill, and gathering payment and shipping information. Resulting orders would be removed from inventory and added to a shipping list (containing shipping information). (10)	(L5)
	c)	Formulate the requirements and challenges for the design of following enterprise application that would integrate the disparate applications within an enterprise into a coherent whole. Also construct a Full Stack architecture that supports Separation of Concerns in this enterprise application which realizing these requirements and challenges Over many years a University has developed a number of separate systems for student administration. There is a system for handling fees in the Accounts department, one for handling registration at the academic Section of College level and yet another for registration within each department. There is a further system to hold examination marks and maintain student academic records, another to handle module entries and yet another to facilitate student feedback concerning courses and lecturers. In those departments that lack technical IT expertise there are still paper-based systems for student feedback and module entry. One result of the lack of an integrated enterprise IT strategy is that	(L6)

students are required to join a series of queues in different buildings at registration and to provide the same information over and over again. The University offers an enterprise computing system that would integrate the disparate applications within that university into a coherent whole. This should be a multi-tier architecture which is flexible in its support of various client programs which will handle interaction with users. (12)

2. a) What is SQL injection? Give its impact and prevention solution from JDBC APIs. (3) (L2)

b) i) Compare and contrast Static SQL and Dynamic SQL with use cases. (5) (L4)

ii) Defend the need for NO SQL in an Enterprise Applications and also compare the role of SQL and NO SQL in Data enabling aspects of Enterprise Applications. (5) (L5)

c) i) 1) Arrange the various components of JDBC application architecture and mention the flow of objects creation and Justify how JDBC increases the longevity of Java enterprise DB applications code over embedded SQL form like SQLJ. (6) (L4)

2) Discuss the features 'Connection Pooling'and 'Distributed Transaction' of JDBC in enterprise Web application (6) (L4)

(OR)

ii) 1) How would you apply what you learned about MVC Architecture in the development of an Enterprise Applications? List its Benefits (6) (L4)

2) How can you assess the value and importance of SPRING and SPRING BOOT framework for the development of an enterprise Applications? (6) (L4)

3 a) Assess the usefulness and values of Java script in all stages of Full Stack based Enterprise Applications (3) (L4)

b) Evaluate the comparative advantages of single-page applications versus multipage applications in web development, with a specific emphasis on their impact on user experience and development efficiency. (10) (L4)

c) Compare and Contrast the use of MERN stack and MEAN Stack for the development of the following Applications (L4)

You have been asked to develop Bookface, a social networking site for people who love reading, and always have their face in a book. The site will allow avid readers to comment on their favourite books, and to read the comments made by others on their favourite books, and other books by their favourite authors. Users must register before they can submit comments on a book. Users should choose a screen name and they should supply an email address. You have been asked to ensure that the site cannot be accessed by webbots and therefore you should ask users to solve a CAPTCHA before they can access the site.

The site should list authors together with their books. Comments are made on books, not on authors. Users can add authors if they are not already there, and they can add books to authors (12)

4. a) Compare and Contrast APIs and Micro Services with respect to inter-enterprise applications (3) (L4)

b) Analyse the following Case study for the use of SOAP and REST based Web services (L5)

Consider an online cab service management system; Cab Service Company employees are categorized into sections. Each section has a section manager. Section No and a fixed basic salary rate is assigned to a section. They keep data on Emp-no, NIC No, name, address, sex, DOB, contact no. of employees. If the employee is a driver his driving license no and capable vehicle classes should be added. Each employee is assigned only to one section. For a travel, driver is assigned to a vehicle. They keep following data about vehicles; Reg No, Insurance no, Model, vehicle category, current mileage. In this case customer is the person who is making a reservation. They keep name, NIC_no, address, contact no. regarding the customer. Once a customer reserved a cab, he cannot make any other reservations until the end of the reserved duration. Customer can reserve one or more vehicles from a reservation. They are keeping index no, payment method, amount, No. of vehicles, distance, start date and time, end date and time regarding a reservation. They are supplying various kinds of supplies to maintain vehicles. A supplier can supply more than one supplying category. Company always keeps data on additional suppliers. Every supplier has a Reg No, Name, address, contact no. A employee is responsible for one or more supplying categories. They keep data like cat_no, cat_description, quantity, net cost regarding every supplying category. To ensure that the site cannot be accessed by webbots and therefore you should ask users to solve a CAPTCHA before they can access the site. (10)

c) i) With illustrations, outline the various step by step tasks with syntax and semantics for the Design, Creation, Deploying, Testing and Consuming REST Web services for a Travel agency website like booking.com, yatra.com etc. (12) (L5)

(OR)

ii) 1) Analyse and discuss the various Standard stack used in SOAP based Web Services in realizing the programming language, Network Protocol and Platform agnostic. (6) (L5)

2) Analyse all Architectural Constraints with their benefits in REST architecture of any enterprise applications (6) (L5)

CO : COURSE OUTCOME; BTL : BLOOM'S TAXONOMY LEVEL

/END/

FD/JU